NexusGrid Technologies

Whitepaper: Distributed Sodium-Ion Microgrids (DSIM) for Urban Resilience

1. Executive Summary NexusGrid Technologies has developed a commercial-scale Distributed Sodium-Ion Microgrid (DSIM) architecture designed to replace legacy lithium-ion grid storage. By utilizing abundant sodium precursors and eliminating conflict minerals like cobalt and nickel, DSIM achieves a 40% reduction in Levelized Cost of Storage (LCOS) while maintaining an energy density of 160 Wh/kg. The technology is currently at Technical Readiness Level (TRL) 7, with a fully operational 500 kWh pilot deployed in a controlled urban grid environment.

2. Core Electrochemistry & Feasibility Unlike theoretical quantum or zero-point energy models, NexusGrid relies on established hard-carbon anode and Prussian blue analogue (PBA) cathode electrochemistry. The innovation lies in our proprietary polymer-ceramic composite solid electrolyte, which suppresses sodium dendrite formation at high current densities. Lab testing confirms 8,000 stable charge-discharge cycles with less than 12% capacity degradation. The thermodynamic limits are strictly adhered to, ensuring high Coulombic efficiency (99.2%) without violating energy conservation principles.

3. Software Architecture & Battery Management System (BMS) To manage the decentralized microgrid, NexusGrid utilizes the EdgeCore BMS. Each battery rack is equipped with wireless IoT sensors that monitor thermal expansion, voltage, and State of Charge (SoC) every 50ms.

- **Network Topology:** The racks communicate via a local wireless Zigbee mesh network, which aggregates data at a central gateway before transmitting it to a cloud-based AWS dashboard via 5G cellular modules.
- **Maintenance:** To ensure zero downtime, the EdgeCore BMS supports Over-The-Air (OTA) firmware updates. Updates are pushed globally from the central server to all active microgrids simultaneously.

4. ESG Impact & Supply Chain The DSIM system represents a massive leap in ESG compliance. By utilizing sodium extracted from seawater desalination brine and synthesizing hard carbon from agricultural waste, NexusGrid completely bypasses the geopolitical and humanitarian risks associated with the lithium and cobalt supply chains.

5. Capital Requirements & Smart Milestone NexusGrid is raising a $6.2M Seed Round to scale manufacturing.

- **Target Escrow Milestone:** Capital will be unlocked upon the "Successful continuous operation of a 1 MWh commercial pilot facility for 72 hours, demonstrating a sub-10ms load-balancing response time verified by a third-party grid operator."